

Notes: Roberts & Sufi (JFE, 2009)

Renegotiation of Financial Contracts: Evidence from Private Credit Agreements

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1 Big picture

- **Question.** How often are private debt contracts renegotiated before maturity, what changes in renegotiation, and what determines the incidence and outcome of renegotiation?
- **Empirical setting.** The paper studies 1,000 private credit agreements between U.S. public non-financial firms and financial institutions from 1996 to 2005.
- **Data innovation.** The authors hand collect renegotiations from borrower SEC filings and connect them to Compustat, CRSP, Dealscan, and original contract terms.
- **Definition of renegotiation.** A renegotiation is the first ex post change before stated maturity to a material loan term: principal amount, interest spread, or maturity.
- **Main frequency result.** Renegotiation is the norm. Unconditionally, 64.5% of all loans are renegotiated. Among loans with maturities above one year, over 90% are renegotiated; among loans above three years, over 96% are renegotiated.
- **Main timing result.** Renegotiations occur early in the life of the loan rather than at maturity. Renegotiated loans are modified after 538 days on average, about 43.6% of stated maturity in Table 1.
- **Main outcome result.** Renegotiations produce large changes: about 766 days in maturity, \$193 million in loan amount, and 64 basis points in spread in absolute value.
- **Not mostly distress.** Covenant violations matter for borrower-unfavorable renegotiations, but most renegotiations are not directly tied to covenant violations or payment defaults.
- **Determinants.** Changes in borrower credit quality, investment opportunities, collateral, and outside financing options drive renegotiation. Macroeconomic and bank-credit conditions also matter.
- **Contract-design result.** Initial contract terms do not mainly prevent renegotiation. Instead, contingencies such as pricing grids and covenants shape the bargaining game by changing parties' default options in specific states.
- **Economic takeaway.** Private debt contracts are deliberately incomplete and routinely rewritten. Contract design is less about avoiding renegotiation and more about allocating bargaining power when renegotiation inevitably occurs.

2 Data and measurement (credit agreements, renegotiations, and contract terms)

2.1 Why private credit agreements are the right laboratory

- Private credit agreements are concentrated debt contracts, which reduces collective-action problems relative to public bonds.
- Bank loans and private credit agreements are a major source of corporate external finance.
- The setting is close to the contracting theory environment: borrowers and lenders write detailed contracts, learn new information, and can renegotiate ex post.

Interpretation. If renegotiation matters anywhere in corporate finance, it should matter in private credit agreements. The paper therefore studies a setting where renegotiation is feasible and economically important.

2.2 Sample construction

- The starting point is a random subsample of 1,000 private credit agreements from the 3,720 SEC-filed agreements collected by Nini, Smith, and Sufi.

- The sample covers U.S. publicly traded non-financial borrowers from 1996 to 2005.
- The authors match the agreements to:
 - Compustat for accounting data;
 - CRSP for stock returns;
 - Dealscan for loan origination terms and some renegotiated terms;
 - SEC 10-Q, 10-K, and 8-K filings for renegotiation details.

Interpretation. The data are unusually detailed because the authors track the actual debt contract through subsequent SEC filings.

2.3 Definition of renegotiation

- A renegotiation is recorded when the borrower reports an ex post change in one of three material terms:
 - loan principal amount or borrowing capacity;
 - interest-rate spread;
 - maturity.
- The renegotiation must occur after origination but before the stated maturity date.
- Changes mechanically dictated by the initial contract are not counted as renegotiations.
- Pure covenant amendments are not counted unless they change principal, spread, or maturity.
- Only the initial renegotiation is collected for each contract.

Interpretation. The definition matches the contracting literature: renegotiation means rewriting material contract terms after information or circumstances change.

2.4 Loan outcomes

Each loan can have one of four mutually exclusive outcomes:

- renegotiated before stated maturity;
- matured without renegotiation;
- disappeared from the SEC filing sample;
- still active at the end of the sample.

Interpretation. Disappeared and end-of-sample loans are right-censored outcomes. The paper handles them carefully because otherwise one could understate the true frequency of renegotiation.

2.5 Renegotiation outcomes

The paper classifies renegotiations into outcome categories:

- **Borrower favorable:** amount increases without a spread increase, or spread decreases without an amount decrease.
- **Borrower unfavorable:** amount decreases without a spread decrease, or spread increases without an amount increase.

- **Amount increasing, not favorable:** amount increases but the spread also increases or the spread change is ambiguous.
- **Amount decreasing, not unfavorable:** amount decreases but the spread also decreases or the spread change is ambiguous.
- **Maturity-only changes:** only maturity increases or decreases.

Interpretation. These labels are relative to the initial contract. All observed renegotiations must be mutually beneficial relative to the parties' contractual default options at the time they occur.

2.6 Borrower characteristics

The main borrower variables are measured from Compustat and CRSP:

- log assets and asset growth;
- debt-to-EBITDA and book leverage;
- market-to-book ratio;
- EBITDA/assets and EBITDA volatility;
- equity returns;
- credit ratings.

Interpretation. These variables proxy for collateral, liquidity, credit quality, profitability, investment opportunities, and the borrower's outside financing options.

2.7 Macroeconomic and lender-condition variables

The paper also studies:

- the BB-AAA public bond credit spread;
- aggregate commercial bank leverage;
- GDP growth;
- CRSP value-weighted market return.

Interpretation. Renegotiation is not only about borrower-level information. It also depends on the health of lenders and the availability of outside capital market alternatives.

2.8 Initial contract features

Important ex ante contract terms include:

- stated maturity;
- initial interest spread;
- number of lenders;
- loan amount/assets;
- whether the contract includes a term loan;
- pricing grids;

- borrowing bases;
- financial covenants.

Interpretation. The paper asks whether these contract terms reduce renegotiation or instead shape what happens once renegotiation occurs.

3 Models and empirical specifications

3.1 Renegotiation as surplus creation

The paper’s conceptual logic can be written as:

new information or changed outside options $\Rightarrow$ surplus under a new contract $\Rightarrow$ renegotiation.

Interpretation. If the original contract becomes inefficient or misaligned with new credit quality, borrower needs, or market conditions, both parties may prefer to rewrite it.

3.2 Binary model for whether renegotiation occurs

The paper estimates a probit model:

$$Pr(Renegotiate_{\ell t} = 1) = \Phi(X_{\ell t}\beta), \quad (1)$$

where Φ is the standard normal cumulative distribution function, $X_{\ell t}$ contains borrower, loan, and macro variables, and β is estimated by maximum likelihood.

Interpretation. The model asks which variables increase the probability that loan ℓ is renegotiated in quarter t .

3.3 Positive and negative changes

For a changing variable x , the paper decomposes changes into:

$$\Delta x^+ = \max(\Delta x, 0), \quad \Delta x^- = \min(\Delta x, 0). \quad (2)$$

Interpretation. This avoids imposing symmetry. A deterioration in credit quality need not have the same effect, with the opposite sign, as an improvement.

3.4 Marginal effects

For the probit model, the table reports marginal effects as the difference in predicted probabilities when a covariate moves one standard deviation above versus below its mean, holding other variables at their means.

Interpretation. The marginal effects are easier to interpret than the raw probit coefficients because they translate estimates into changes in renegotiation probability.

3.5 Multinomial logit for renegotiation outcomes

For outcome categories $j = 1, \dots, 5$, the paper estimates:

$$Pr(Y_{\ell t} = j) = \frac{\exp(\beta_j X_{\ell t})}{\sum_{k=1}^5 \exp(\beta_k X_{\ell t})}. \quad (3)$$

The baseline category is no renegotiation or maturity-only renegotiation. The main outcome categories are borrower favorable, amount increase, amount decrease, and borrower unfavorable.

Interpretation. The multinomial model asks not just whether a renegotiation occurs, but who gains contractually and how the loan changes.

3.6 Early versus late renegotiations

The paper defines:

$$Late_{\ell t} = 1$$

if at least three years have elapsed since loan origination, and zero otherwise. It then interacts this indicator with ex post changes in borrower and macro variables.

Interpretation. This asks whether fast-moving variables and slow-moving variables matter differently depending on how much time has passed since origination.

3.7 Contract contingency interaction tests

The paper estimates regressions that interact a contract contingency with the variable on which the contingency is written. Examples include:

$$PricingGrid_{\ell}^{cashflow} \times \Delta(EBITDA/Assets)_{\ell t}$$

and

$$Covenant_{\ell}^{networth} \times \Delta(NetWorth/Assets)_{\ell t}.$$

Interpretation. If contingencies allocate bargaining power, then renegotiation outcomes should become more sensitive to the underlying state variable when the original contract includes a contingency tied to that variable.

4 Identification and empirical design

4.1 What identifies the main effects

- The analysis uses variation across loans and over time in borrower characteristics, macro conditions, and initial contract design.
- The paper studies first renegotiations, which avoids repeated-event dependence but limits information about later amendments.
- It controls for industry, loan purpose, credit ratings, firm characteristics at origination, deal characteristics, and time trends.
- Standard errors are robust to within-firm dependence and heteroskedasticity.

Interpretation. This is an observational design. It is not a randomized experiment, but the rich contract and borrower data allow the authors to connect renegotiation to changing states of the world.

4.2 Why SEC filings matter

- SEC filings disclose material debt agreements and material modifications.
- This allows the authors to observe renegotiations that may appear in Dealscan as new loans, even though economically they are amendments of prior agreements.
- The authors find that 47% of renegotiations generate independent Dealscan observations.

Interpretation. A major contribution is showing that many loans observed as new in Dealscan are actually renegotiations of existing contracts.

4.3 Addressing distress-only interpretations

- The paper compares renegotiations to covenant violations in the year before, quarter of, and year after renegotiation.
- Covenant violations are strongly related to unfavorable and amount-decreasing renegotiations.
- But covenant violations are rare around favorable and amount-increasing renegotiations.
- Most renegotiations therefore occur outside distress or default.

Interpretation. The paper broadens renegotiation beyond the default literature. Contract rewriting is a normal feature of private debt, not merely a distress response.

4.4 Addressing contract-design endogeneity

- A simple interpretation would be that contingencies such as pricing grids reduce the need for renegotiation.
- The paper finds the opposite pattern: contracts with contingencies are often more likely to be renegotiated.
- The authors acknowledge endogeneity: contingencies may be placed into contracts expected to need renegotiation.
- Because renegotiation is near universal in long-term loans, there is limited scope for contingencies to reduce renegotiation incidence.

Interpretation. The more persuasive interpretation is that contingencies determine bargaining positions and default options in renegotiation.

4.5 Why outcome heterogeneity matters

- A contract change can increase credit availability, reduce credit availability, lower spreads, raise spreads, or change maturity.
- A single renegotiation dummy would hide this heterogeneity.
- The multinomial outcome analysis shows that different shocks move different renegotiation outcomes.

Interpretation. The paper's core contribution is not only measuring renegotiation frequency. It also shows how renegotiation reallocates credit, price, maturity, and bargaining power.

4.6 What the paper cannot fully prove

- Initial contract terms are endogenous.
- Borrower and lender conditions are not randomly assigned.
- Some interest-spread changes are missing when renegotiations are identified only from SEC text and not Dealscan.
- The data include only the first renegotiation.
- The sample focuses on material credit agreements of public firms, not all private loans.

Interpretation. The evidence is strongest as a large-scale descriptive and conditional analysis of renegotiation. Causal claims about individual contract features require caution.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Adjacent figures and tables are grouped compactly when possible.

5.1 Table 1 and Figure 1: frequency and timing of renegotiation

Read:

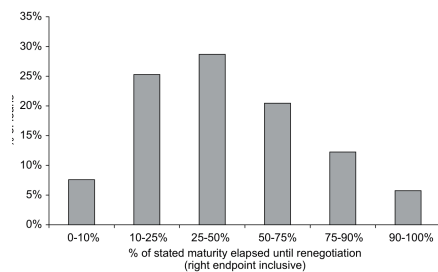
- Table 1 shows that 64.5% of the 1,000 loans are renegotiated before stated maturity.
- 20.8% mature without renegotiation, 9.5% disappear from the sample, and 5.2% are still active at the end of the sample.
- Renegotiated loans are modified after 538 days on average, versus an average stated maturity of 1,367 days.
- The renegotiation occurs after 43.6% of stated maturity has elapsed.
- Conditional on maturity above one year, renegotiation exceeds 90%; conditional on maturity above three years, it exceeds 96%.
- Figure 1 shows that the mass of renegotiations occurs well before maturity, especially between 25% and 50% of stated maturity.

Economic message. Renegotiation is not an exceptional event. For long-term private credit agreements, it is close to inevitable.

Panel A: Loan outcomes						
Loan outcome	Fraction of low loans	Merge or acq.	Go private	Loan duration (days)	Stated maturity (days)	Loan duration/stated maturity
Renegotiated	0.645	0.020	0.000	538	1,367	0.436
Matured	0.208	0.024	0.000	506	506	1.000
Right censored outcomes						
Disappeared	0.095	0.589	0.021	453	1,463	0.324
End of sample	0.052	0.000	0.000	784	1,747	0.459
Loans	1,000	1,000	1,000	1,000	1,000	1,000

Panel B: Loan outcomes by maturity				
	Stated maturity bin (years)			
	<1	1-3	3-5	>5
Renegotiated	0.268	0.727	0.941	0.983
Matured	0.732	0.273	0.059	0.017
Loans	194	172	255	232

(a) Table 1: loan outcomes



(b) Figure 1: when renegotiation occurs

5.2 Tables 2–3: borrower and contract characteristics

Read:

- Table 2 shows that sample borrowers are larger than the typical Compustat firm: mean assets are \$3.36 billion versus \$1.93 billion.
- Sample borrowers have higher debt-to-EBITDA than Compustat firms, but lower cash/assets and lower EBITDA volatility.
- Almost half of the sample borrowers have an S&P credit rating, compared with about 12% of Compustat firms.
- Table 3 shows that the average contract amount is \$450 million, the average spread is 162 bps over LIBOR, and average maturity is 1,217 days.
- 72.6% of agreements include a pricing grid; 19.4% include a borrowing base; 95.3% include financial covenants.

Economic message. The sample consists of large, material credit agreements with highly detailed contingencies. These are exactly the kinds of contracts where renegotiation and bargaining power should matter.

Panel A: Firm characteristics						
	Renegotiation sample			Compustat sample		
	Mean	SD	Median	Mean	SD	Median
Assets (\$Mil)	3.360	10.792	751	1,927	9,706	103
Debt/EBITDA	10.728	20.069	7.502	5.742	24.533	1.958
Book leverage	0.295	0.195	0.277	0.330	0.497	0.226
EBITDA/Assets	0.035	0.024	0.034	-0.049	0.308	0.024
Market-to-book ratio	1.734	0.986	1.424	4.026	10.199	1.600
Asset tangibility	0.339	0.232	0.286	0.293	0.248	0.212
Cash/Assets	0.076	0.099	0.038	0.187	0.231	0.082
Altman Z-Score	2.582	2.839	1.803	2.405	23.053	1.800
S&P Credit rating	0.496	0.500	0.000	0.119	0.324	0.000
EBITDA Volatility	0.019	0.021	0.012	0.063	0.155	0.021
Firm-quarter obs	6,813	6,813	6,813	297,423	297,423	297,423
Firms	799	799	799	16,952	16,952	16,952

Panel B: Distribution of firms across industries						
	Renegotiation sample			Compustat sample		
	Mean	SD	Median	Mean	SD	Median
Consumer non-durables			0.084			0.061
Consumer durables			0.034			0.028
Manufacturing			0.160			0.110
Oil, gas, & coal			0.059			0.053
Chemicals & allied products			0.030			0.024
Business equipment			0.113			0.237
Telephone & television trans.			0.056			0.054
Utilities			0.059			0.031
Wholesale, retails, & some srvs.			0.179			0.117
Healthcare, medical equip., & drugs			0.077			0.113
Everything else			0.150			0.171
Firm-quarter obs			6,813			297,423
Firms			799			16,952

Figure 2: Table 2: borrower summary statistics

	Renegotiation sample			Dealscan sample		
	Mean	SD	Median	Mean	SD	Median
Contract terms						
Amount (\$Mil)	450	1,122	185	311	739	105
Spread over LIBOR (bps)	162	114	138	206	137	200
Number of lenders	8,603	8,187	6,000	6,000	7,574	3,000
Term loan in deal	0.262	0.440	0.000	0.339	0.473	0.000
Stated maturity (days)	1,217	643	1,096	1,283	6,672	1,098
Contract contingencies						
Pricing grid	0.726	0.446	1.000			
Any measure of cash flow	0.367	0.482	0.000			
Credit rating	0.232	0.422	0.000			
Balance sheet item	0.049	0.216	0.000			
Other measure	0.078	0.268	0.000			
Borrowing base	0.194	0.396	0.000			
Accounts receivable	0.166	0.372	0.000			
Inventory	0.134	0.341	0.000			
Other collateral	0.036	0.186	0.000			
Financial covenant	0.953	0.212	1.000			
Any measure of cash flow	0.826	0.379	1.000			
Any measure of net worth	0.449	0.498	0.000			
Balance sheet item	0.318	0.466	0.000			
Measures of liquidity	0.141	0.348	0.000			
Loans	1,000	1,000	1,000	25,326	25,326	25,326
Firms	799	799	799	12,389	12,389	12,389

Figure 3: Table 3: contract summary statistics

5.3 Table 4 and Figure 2: what changes in renegotiation, and when

Read:

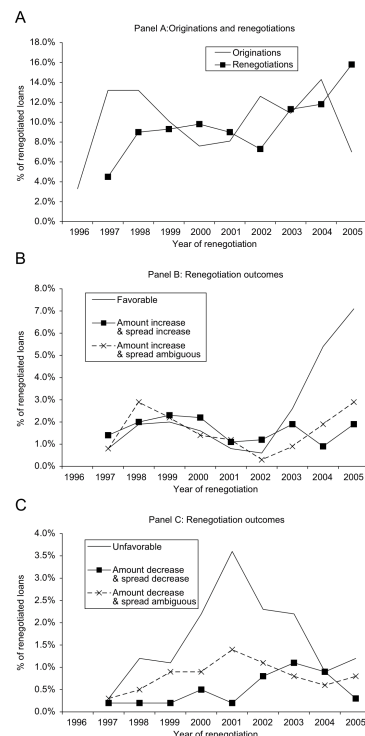
- Table 4 Panel A shows large absolute changes: 766 days of maturity, \$193 million in loan amount, and 64 bps in spread.
- Relative to initial terms, these are roughly 64% of maturity, 43% of amount, and 40% of spread.
- Panel B shows that 56.0% of renegotiations increase the amount, 29.1% decrease it, 56.9% extend maturity, and 11.9% shorten maturity.
- 28.4% are borrower favorable, while 15.2% are borrower unfavorable.
- Figure 2 shows cyclicity: borrower-favorable renegotiations rise late in the sample, while unfavorable outcomes spike around 2001.

Economic message. Renegotiation materially changes corporate capital structure. It is not a small waiver or administrative amendment.

Panel A: Changes in absolute value of contract terms due to renegotiation					
Variable	Obs	Mean	SD	Min	Max
[Change in maturity (days)]	296	766	528	0	2,093
[Change in amount (\$Mil)]	301	193	548	0	8,400
[Change in spread (bps)]	301	64	78	0	575

Panel B: Likelihood and timing of renegotiation outcomes conditional on renegotiation		
	Likelihood of outcome	Loan duration/stated maturity
<i>Renegotiation outcomes</i>		
Amount increase	0.560	0.416
Amount decrease	0.291	0.446
Spread increase	0.293	0.460
Spread decrease	0.259	0.486
Maturity increase	0.569	0.521
Maturity decrease	0.119	0.346
Switch lead arranger	0.114	0.528
<i>Mutually exclusive outcomes</i>		
Borrower favorable	0.284	0.455
Amount increase, not favorable	0.336	0.392
Amount increase, spread increased	0.161	0.426
Amount increase, spread change ambiguous	0.175	0.360
Amount decrease, not unfavorable	0.181	0.423
Amount decrease, spread decreased	0.048	0.493
Amount decreased, spread change ambiguous	0.079	0.408
Fraction of deal amount prepaid	0.012	0.341
Deal terminated	0.042	0.396
Borrower unfavorable	0.152	0.496
Only maturity increased	0.045	0.499
Only maturity decreased	0.002	0.177
Firm-quarter obs	645	645

(a) Table 4: renegotiation summary statistics



(b) Figure 2: calendar-year renegotiation patterns

5.4 Table 5: covenant violations and renegotiation

Read:

- Borrower-favorable renegotiations are rarely preceded by covenant violations: 4.9% have a violation in the prior year.
- Amount-increasing renegotiations are also rarely tied to prior-year violations: 4.1%.
- Amount-decreasing renegotiations are more distress-related: 17.1% have a covenant violation in the prior year.
- Borrower-unfavorable renegotiations have the strongest distress connection: 21.4% have a prior-year violation, 12.2% a contemporaneous violation, and 14.3% a next-year violation.

Economic message. Covenant violations matter for unfavorable renegotiations, but they do not explain the bulk of renegotiation activity. Most renegotiations happen outside formal default.

Renegotiation outcome	Obs	Last year	Contemporaneous	Next year
Borrower favorable	183	0.049	0.016	0.022
Amount increase	217	0.041	0.032	0.042
Amount decrease	117	0.171	0.034	0.052
Borrower unfavorable	98	0.214	0.122	0.143

Figure 5: Table 5: renegotiation and covenant violations

5.5 Table 6: determinants of whether renegotiation occurs

Read:

- Panel A shows that positive asset growth raises renegotiation probability. A one-standard-deviation positive change in log assets has a 1.274% marginal effect.
- Positive book-leverage changes and equity returns also raise renegotiation probability.
- Negative changes in EBITDA/assets and equity returns are also important, showing that both good and bad news can trigger renegotiation.
- A widening credit spread raises renegotiation likelihood, and changes in bank leverage matter, especially on the negative-change side.
- Panel B shows that the initial stated maturity is a strong predictor of renegotiation.
- Pricing grids and borrowing bases do not reduce renegotiation incidence; covenant-on-cash-flow and young-loan indicators are positively related to renegotiation.

Economic message. Renegotiation is triggered by new information about the borrower and by shifts in credit-market conditions. Initial contingencies are not mainly devices that avoid renegotiation.

Panel A: Ex post change determinants (loan-quarter panel sample)				
	Positive change		Negative change	
	Coefficient	Marginal effect (%)	Coefficient	Marginal effect (%)
<i>Change in firm characteristics</i>				
ΔLog assets	0.405* (2.256)	1.274*	0.076 (0.158)	0.062
ΔDebt/EBITDA	0.001 (0.399)	0.147	-0.003 (-1.756)	-0.560
ΔBook leverage	1.271** (2.639)	1.041**	0.190 (0.289)	0.123
ΔMarket-to-book	-0.218 (-1.633)	-0.814	0.012 (0.123)	0.082
ΔEBITDA/Assets	6.493 (1.718)	0.706	-6.256* (-2.266)	-0.918*
ΔEBITDA Volatility	-9.622 (-1.570)	-0.647	-9.735 (-1.613)	-0.742
Equity return	0.173** (2.668)	1.286**	-0.579* (-2.554)	-1.174*
<i>Change in macroeconomic factors</i>				
ΔCredit spread	0.142* (2.531)	1.021*	-0.023 (-0.454)	-0.210
ΔBank leverage	171.234* (2.035)	0.711*	-43.439** (-4.554)	-2.124**
ΔGDP Growth	9.531 (1.130)	0.451	-4.562 (-0.692)	-0.259
ΔStock market return	0.595 (1.639)	0.645	-0.066 (-0.170)	-0.068
<i>Control variables</i>				
Industry fixed effects	Yes			
Deal purpose fixed effects	Yes			
Credit ratings fixed effects	Yes			
Firm characteristics at origination	Yes			
Deal characteristics	Yes			
Time trend	Yes			
Pr (Renegotiation) (%)	9.131	-	-	-
Renegotiations	513	-	-	-
Firms	760	-	-	-
Loans	936	-	-	-
Obs	5,618	-	-	-
Pseudo-R ²	0.088	-	-	-

Figure 6: Table 6 Panel A: ex post determinants of renegotiation

Table 6 (continued)				
Panel B: Ex ante determinants (loan cross-sectional sample)				
	With ex post changes		Without ex post changes	
	Coefficient	Marginal effect (%)	Coefficient	Marginal effect (%)
<i>Firm characteristics at origination</i>				
Log assets	0.077	4.189	0.080	4.772
Debt/EBITDA	0.001	0.770	-0.001	-0.730
Book leverage	-0.326	-2.040	-0.039	-0.267
Market-to-book	0.108	3.898	-0.002	-0.084
EBITDA/Assets	-1.890	-1.586	-0.891	-0.810
EBITDA Volatility	3.052	2.081	1.920	1.436
<i>Deal characteristics</i>				
Ln (Stated maturity)	1.013**	23.968**	0.767**	19.477**
IR Spread	0.000	0.341	0.001	2.180
Number of lenders	-0.013	-3.480	-0.006	-1.654
Loan amount/Assets	-0.104	-1.097	0.064	0.715
Term loan in deal	0.158	2.174	0.133	2.077
Pricing grid	0.048	0.717	0.033	0.542
Borrowing base	0.325	4.078	0.316*	4.289*
Covenant on cash flow	0.630**	8.729**	0.569**	8.240**
Young loan	0.310*	5.053*	0.309**	5.581**
<i>Control variables</i>				
Industry fixed effects	Yes		Yes	
Deal purpose fixed effects	Yes		Yes	
Credit ratings fixed effects	Yes		Yes	
Change in firm characteristics	Yes		No	
Change in macroeconomic factors	Yes		No	
Time trend	Yes		Yes	
Pr(Renegotiation) (%)	67.896	-	64.586	-
Renegotiations	497	-	600	-
Firms	622	-	751	-
Loans	732	-	929	-
Obs	732	-	929	-
Pseudo-R ²	0.310	-	0.194	-

Figure 7: Table 6 Panel B: ex ante determinants of renegotiation

5.6 Table 7: determinants of renegotiation outcomes

Read:

- Panel A shows that positive changes in log assets raise the probability of favorable and amount-increasing renegotiations.
- Deteriorating debt-to-EBITDA lowers the probability of a favorable outcome.
- Higher market-to-book ratios reduce borrower-unfavorable and amount-decreasing outcomes, consistent with better outside financing options.
- Changes in bank financial health affect several outcomes, suggesting lender balance sheets shape bargaining.
- Panel B shows that initial maturity matters for favorable, amount-increasing, amount-decreasing, and unfavorable outcomes.
- Covenants on cash flow are associated with several types of renegotiation outcomes, consistent with contract features shaping bargaining.

Economic message. The outcome of renegotiation depends on how borrower quality and outside options change. Lenders respond more favorably to collateral and realized improvements than to growth opportunities alone.

standard errors robust to within-firm dependence and heteroskedasticity.

Panel A: Marginal effects of ex post determinants (figures in %)				
	Favorable	Amount increase	Amount decrease	Unfavorable
<i>Change in firm characteristics</i>				
ΔLog assets	0.834*	1.269**	0.221	0.011
	(2.060)	(4.092)	(0.762)	(0.549)
ΔDebt/EBITDA	-0.700*	0.074	-0.128	0.012
	(-2.483)	(0.234)	(-0.384)	(0.775)
ΔBook leverage	0.452	0.329	0.021	0.052**
	(1.357)	(1.358)	(0.116)	(3.097)
ΔMarket-to-book	-0.738	0.224	-0.857*	-0.076*
	(-1.406)	(0.475)	(-2.044)	(-2.219)
ΔEBITDA/Assets	0.948	-0.071	-0.271	-0.059**
	(1.864)	(-0.243)	(-1.141)	(-3.458)
ΔEBITDA Volatility	-0.531	-0.433	-0.225	-0.002
	(-1.514)	(-1.659)	(-0.892)	(-0.142)
Equity return	1.170**	0.010	0.151	-0.034
	(3.182)	(0.089)	(0.528)	(-0.822)
<i>Change in macroeconomic factors</i>				
ΔCredit spread	0.106	0.143	0.411	-0.000
	(0.343)	(0.457)	(1.145)	(0.013)
ΔBank leverage	-1.562**	-0.237	-0.430	-0.045*
	(-4.310)	(-0.769)	(-1.348)	(-2.350)
ΔGDP Growth	0.189	0.631*	-0.293	-0.031
	(0.625)	(2.259)	(-0.985)	(-1.897)
ΔStock market return	0.677*	-0.131	0.065	0.025
	(1.975)	(-0.418)	(0.244)	(1.469)
<i>Control variables</i>				
Industry fixed effects	Yes			
Deal purpose fixed effects	Yes			
Credit ratings fixed effects	Yes			
Firm characteristics at origination	Yes			
Deal characteristics	Yes			
Time trend	Yes			
Outcome probability (%)	2.397	3.036	1.686	1.651
Renegotiations	135	171	95	93
Firms	763	-	-	-
Loans	939	-	-	-
Obs	5,633	-	-	-
Pseudo-R ²	0.123	-	-	-

Figure 8: Table 7 Panel A: ex post determinants of renegotiation outcomes

Table 7 (continued)				
Panel B: Marginal effects of ex ante determinants (figures in %)				
	Favorable	Amount increase	Amount decrease	Unfavorable
<i>Firm characteristics at origination</i>				
Log assets	-0.238	-7.295	21.961**	0.248
			-1.241	0.096
Debt/EBITDA	-1.778	-3.260		
Book leverage	-2.522	11.310	-10.498	-0.278
Market-to-Book	1.557	5.448	1.005	-1.419*
EBITDA/Assets	7.047	2.950	-10.425	-0.306
EBITDA volatility	4.569	-0.465	1.280	0.255
Ln (Stated maturity)	16.197**	22.195**	7.887**	2.195**
IR Spread	7.238	-17.643*	10.853	-0.402
Number of lenders	0.659	-5.291	-4.492	-0.064
<i>Deal characteristics</i>				
Loan amount/Assets	-23.805**	-0.230	16.318*	0.002
Term loan in deal	1.334	5.517	1.147	0.179
Pricing grid	1.603	2.082	0.425	-0.183
Borrowing base	3.223	1.915	6.949	-0.205
Covenant on cash flow	8.834**	3.507*	7.222**	0.255*
Young loan	16.438**	-2.604	3.925	-0.449
<i>Control variables</i>				
Industry fixed effects	Yes			
Deal purpose fixed effects	Yes			
Credit ratings fixed effects	Yes			
Change in firm characteristics	Yes			
Change in macroeconomic factors	Yes			
Time trend	Yes			
Outcome probability (%)	17.976	22.770	12.650	12.383
Renegotiations	135	171	95	93
Firms	640	-	-	-
Loans	751	-	-	-
Obs	751	-	-	-
Pseudo-R ²	0.257	-	-	-

Figure 9: Table 7 Panel B: ex ante determinants of renegotiation outcomes

5.7 Table 8: early versus late renegotiations

Read:

- Table 8 distinguishes early renegotiations from renegotiations after at least three years.
- Panel A shows that declines in debt-to-EBITDA and EBITDA volatility increase renegotiation likelihood in the short run, but the interaction terms reverse part of this relation for late renegotiations.
- Changes in bank leverage are amplified in later renegotiations.
- Panel B shows that outcome determinants also depend on whether renegotiation occurs early or late.
- GDP growth effects appear concentrated in long-run amount-increasing outcomes.

Economic message. Fast-moving borrower shocks matter early, while slower-moving macro and bank-credit conditions matter more after more time has passed since origination.

Panel A: Binomial probit of renegotiation (estimated coefficients)

	Positive change		Negative change	
	Level coefficient	Interaction coefficient	Level coefficient	Interaction coefficient
<i>Change in firm characteristics</i>				
ΔLog assets	0.472* (2.302)	-0.337 (-0.919)	-0.247 (-0.512)	1.517 (0.968)
ΔDebt/EBITDA	-0.001 (-0.446)	0.001 (0.428)	-0.004* (-2.009)	0.053* (2.234)
ΔBook leverage	1.344* (2.444)	0.776 (0.600)	0.020 (0.028)	-1.822 (-1.025)
ΔMarket-to-book	-0.141 (-1.008)	-0.348 (-0.720)	0.008 (0.070)	0.257 (1.582)
ΔEBITDA/Assets	5.224 (1.285)	1.233 (0.112)	-6.435* (-2.068)	-2.902 (-0.427)
ΔEBITDA Volatility	-9.269 (-1.274)	0.186 (0.011)	-12.935* (-2.111)	30.120** (2.887)
Equity return	0.181* (2.451)	-0.031 (-0.170)	-0.574* (-2.362)	-0.856 (-1.245)
<i>Change in macroeconomic factors</i>				
ΔCredit spread	0.175** (2.781)	0.006 (0.027)	-0.049 (-0.935)	0.168 (0.811)
ΔBank leverage	153.464 (1.816)	104411.331* (2.176)	-40.668** (-4.107)	-90.203* (-2.304)
ΔGDP Growth	11.489 (1.308)	-23.360 (-0.734)	-10.555 (-1.540)	42.179 (1.683)
ΔStock market return	0.624 (1.663)	-0.909 (-0.603)	-0.279 (-0.679)	1.442 (1.042)
<i>Control variables</i>				
Industry fixed effects	Yes			
Deal purpose fixed effects	Yes			
Credit ratings fixed effects	Yes			
Firm characteristics at origination	Yes			
Deal characteristics	Yes			
Time trend	Yes			
Pr(Renegotiation) (%)	9.131	-	-	-
Renegotiations	513	-	-	-
Firms	760	-	-	-
Loans	936	-	-	-
Obs	5,618	-	-	-
Pseudo-R ²	0.096	-	-	-

Figure 10: Table 8 Panel A: early versus late renegotiation incidence

Table 8 (continued)

Panel B: Multinomial logit of renegotiation outcomes (estimated coefficients)

	Favorable		Amount increase		Amount increase		Unfavorable	
	Level coefficient	Interaction coefficient	Level coefficient	Interaction coefficient	Level coefficient	Interaction coefficient	Level coefficient	Interaction coefficient
<i>Change in firm characteristics</i>								
ΔLog assets	1.538** (2.763)	-1.383 (-1.133)	1.926** (4.108)	-0.997 (-1.366)	0.057 (0.077)	0.701 (0.700)	0.320 (0.364)	-0.749 (-0.542)
ΔDebt/EBITDA	-0.012** (-2.795)	0.019** (2.911)	0.002 (0.360)	-0.006 (-1.069)	-0.007 (-0.997)	0.021* (2.094)	-0.002 (-0.293)	0.015 (1.659)
ΔBook leverage	1.492 (1.102)	-1.844 (-0.651)	2.043 (1.764)	-5.372 (-1.881)	-0.565 (-0.326)	4.850 (1.940)	5.635** (2.662)	-0.075 (-0.018)
ΔMarket-to-book	-0.356 (-1.174)	-0.265 (-0.622)	0.014 (0.052)	0.160 (0.355)	-0.632 (-1.869)	-0.019 (-0.026)	-1.473** (-2.991)	1.530 (1.948)
ΔEBITDA/Assets	22.884* (2.094)	-22.808 (-1.360)	1.536 (0.207)	-14.218 (-1.035)	-15.475 (-1.922)	27.227 (1.469)	-36.336** (-3.051)	38.054* (2.155)
ΔEBITDA Volatility	-22.325 (-1.524)	46.775 (1.433)	-19.086 (-1.535)	35.117 (1.779)	-22.648 (-1.225)	21.064 (0.794)	-1.578 (-0.076)	-19.885 (-0.639)
ΔEquity return	0.638** (3.051)	-0.160 (-0.343)	0.064 (0.309)	-0.089 (-0.210)	0.037 (0.114)	0.407 (0.606)	-0.008 (-0.013)	-1.491 (-1.546)
<i>Change in macroeconomic factors</i>								
ΔCredit spread	0.095 (0.824)	-0.400 (-0.925)	-0.015 (-0.119)	1.108** (5.397)	0.261 (1.262)	-0.358 (-0.983)	-0.084 (-0.490)	0.596 (1.402)
ΔBank leverage	-145.105** (-4.715)	36.341 (0.555)	-37.467 (-1.135)	-55.017 (-0.938)	-61.586 (-1.348)	-42.773 (-0.509)	-101.129* (-2.389)	-244.866** (-3.778)
ΔGDP Growth	10.285 (0.605)	-18.071 (-0.280)	24.588 (1.446)	117.515** (2.916)	-21.661 (-0.956)	20.967 (0.444)	-35.459 (-1.568)	-29.284 (-0.418)
ΔStock market return	1.900* (2.136)	-0.148 (-0.052)	-0.455 (-0.569)	2.592 (1.323)	0.343 (0.302)	-0.053 (-0.021)	1.487 (1.290)	3.349 (1.158)
<i>Control variables</i>								
Industry fixed effects	Yes							
Deal purpose fixed effects	Yes							
Credit ratings fixed effects	Yes							
Firm characteristics at origination	Yes							
Deal characteristics	Yes							
Time trend	Yes							
Outcome Probability (%)	0.024	-	0.030	-	0.017	-	0.017	-
Renegotiations	135	-	171	-	95	-	93	-
Firms	763	-	-	-	-	-	-	-
Loans	939	-	-	-	-	-	-	-
Obs	5,633	-	-	-	-	-	-	-
Pseudo-R ²	0.133	-	-	-	-	-	-	-

Figure 11: Table 8 Panel B: early versus late renegotiation outcomes

5.8 Table 9 and Figure 3: contingencies shape bargaining power

Read:

- Table 9 interacts contract contingencies with the accounting variable on which the contingency is written.

- Pricing grids on balance sheet or cash-flow measures make borrower-unfavorable outcomes more sensitive to deterioration in those measures.
- Covenants on cash flow or net worth make borrower-favorable outcomes more sensitive to improvements in those measures.
- Figure 3 visualizes the same point. With a pricing grid, low net worth or low cash flow produces a much higher predicted probability of an unfavorable renegotiation.
- With a covenant, positive cash-flow or net-worth changes create a higher predicted probability of a favorable renegotiation.

Economic message. Contingencies do not appear primarily designed to stop renegotiation. They shape the renegotiation game by changing the parties' default option and bargaining power in different states.

DEPENDENT AND INTERDEPENDENCY.

	Favorable	Unfavorable
Model 1		
Pricing grid on balance sheet item	-0.277 (-0.520)	-0.016 (-0.038)
ΔNet worth/Assets	0.000 (0.375)	-0.001* (-2.016)
Interaction	0.000 (0.724)	-0.020** (-3.270)
Model 2		
Pricing grid on cash flow	0.033 (0.162)	-0.152 (-0.581)
ΔEBITDA/Assets	18.310 (1.459)	-8.203 (-0.905)
Interaction	12.993 (0.830)	-56.318** (-3.904)
Model 3		
Covenant on cash flow	0.401 (1.822)	0.547 (1.628)
ΔEBITDA/Assets	-7.890 (-0.697)	-9.163 (-0.496)
Interaction	36.128** (2.696)	-27.161 (-1.401)
Model 4		
Covenant on net worth	-0.014 (-0.068)	-0.077 (-0.335)
ΔNet worth/Assets	-0.000 (-0.190)	-0.002** (-2.588)
Interaction	0.001** (2.689)	0.003* (2.520)
Control variables in each model		
Change in firm characteristics	Yes	
Change in macroeconomic factors	Yes	
Time trend	Yes	

(a) Table 9: contingency interactions

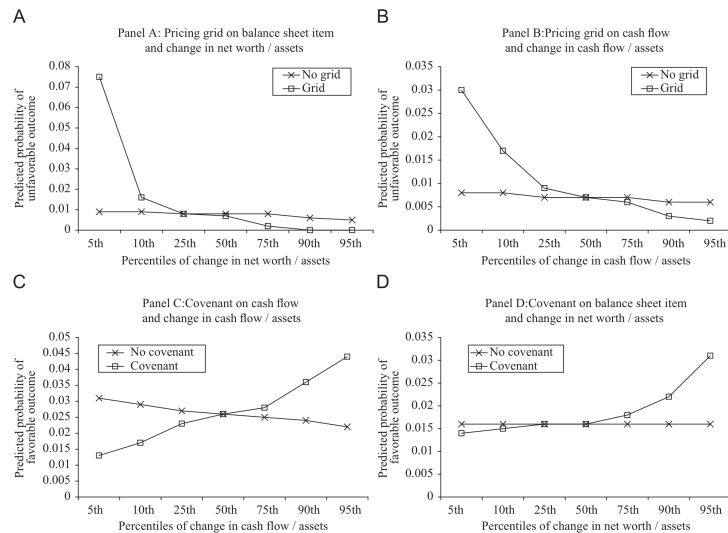


Fig. 3. Changes in firm characteristics and contingencies. The sample consists of 1,000 private credit agreements between financial institutions (lenders) and publicly traded non-financial U.S. companies (borrowers) during the period 1996–2005. Panels A and B show the predicted probability (from the multinomial logit model discussed in Table 8) of an unfavorable renegotiation outcome for two groups of loans as a function of the change in net worth/assets and cash flow/assets over the life of the loan, respectively. The two groups in each panel are loans with and without a performance pricing grid on

(b) Figure 3: predicted probabilities with contingencies

6 Short summary

- The paper studies 1,000 private credit agreements between U.S. public firms and financial institutions over 1996–2005.
- Renegotiation is defined as the first ex post change to loan amount, spread, or maturity before stated maturity.
- 64.5% of loans are renegotiated before maturity.
- Among loans with maturities above one year, renegotiation occurs in more than 90% of cases.
- Renegotiations occur early, typically well before stated maturity.
- Renegotiations change material terms by large amounts: maturity, loan amount, and spread all move substantially.
- Favorable and amount-increasing renegotiations are common.

- Unfavorable and amount-decreasing renegotiations are less frequent but still important.
- Covenant violations are strongly related to unfavorable outcomes but do not explain most renegotiations.
- Borrower asset growth, leverage changes, profitability, equity returns, and investment opportunities predict renegotiation and outcomes.
- Credit-market conditions, bank balance sheets, GDP growth, and stock-market returns also affect renegotiation.
- Initial maturity is a strong predictor of renegotiation.
- Pricing grids, borrowing bases, and covenants do not mainly prevent renegotiation.
- Contingencies make renegotiation outcomes more sensitive to the state variables on which those contingencies are written.
- The paper's strongest contract-design interpretation is that contingencies allocate bargaining power in a state-contingent way.
- The results imply that contract incompleteness and renegotiation are central, not peripheral, to private debt contracting.

7 Conclusion

7.1 Core conclusion

Roberts and Sufi show that private debt renegotiation is pervasive, early, and economically large. Long-term private credit agreements are not static contracts that remain in force until maturity. They are flexible contracting arrangements that are routinely rewritten as borrower quality, investment opportunities, lender health, and macroeconomic conditions change.

7.2 Economic intuition

The intuition is that the initial loan contract is incomplete. Once new information arrives, the old terms may no longer allocate credit, maturity, and pricing efficiently. Renegotiation lets borrower and lender split the surplus from rewriting the agreement. But the split depends on bargaining power, and the initial contract can affect that bargaining power through pricing grids, borrowing bases, and covenants.

7.3 Takeaway

The paper changes how we should read private credit agreements. Their detailed contingencies are not simply mechanisms to avoid costly renegotiation. Because renegotiation is almost inevitable, these terms are best understood as instruments that shape the future renegotiation game.